Balanced-pair accumulation screen: diagnostic report

Question. Does keeping selected points in balanced Eisenstein-pair coordinates through the radix-14 group sum merit a qualified performance comparison against the ordinary-field cached paths?

Scope and method. This is a source and evidence review of the benchmark-only pair-accumulation overlay, not a production change. The fixture contains 12,288 frozen secp256k1 scalar/point pairs; its base fixture SHA-256 is `7f4672b7b676969fe40056d0f14c6ecf45ad36109870cdf3466ebafd9e7b22d8`. The curve is SEC 2 secp256k1 over `p = 2^256 - 2^32 - 977`, with subgroup order `n = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141` and generator `G`; the native provenance archive carries the exact fixture and executable. The hash-guarded overlay prepares the frozen norm-136531 affine table as balanced `(x,y)` pairs, applies unit rotations and signs in the pair field, performs mixed Jacobian additions, batch-normalizes 128 outputs, and converts final affine values back to conventional field elements. The measured path preserves preparation and online time as separate phases.

Evidence. The run is identified by native provenance archive SHA-256 `2f36e7c6d52a85790f17428daedec09c95d3c3e936f44a655c18725ecb2c9c1f`. The independent receipt auditor reproduces `pair-accumulation-audit.json`: status `verified_exact_replay`, 12,288 outputs, 28,825,228 retained bytes, 4,900.576860 ms preparation, and 1,036.845373 ms online. The archive includes the exact source, executable, fixture, build logs, stdout, and stderr. The existing result record links the implementation and provenance.

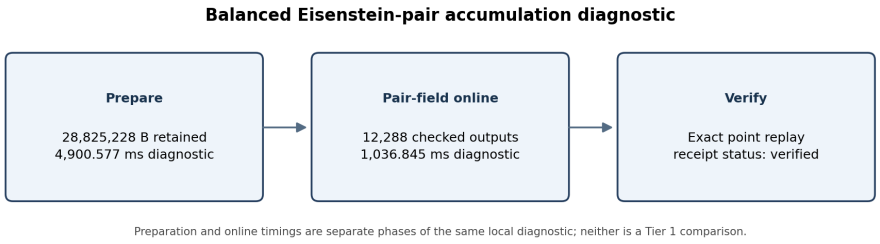


Figure 1: Preparation, pair-field online accumulation, and exact output verification pipeline

Finding and limits. The replay is positive correctness evidence and the performance screen is negative: pair-field online accumulation took about 1.04 seconds after about 4.90 seconds of table preparation, while nearby ordinary-field cached diagnostics were roughly 64–68 ms. Those neighboring timings are not a matched panel, so this evidence has no paired interval and does not establish a qualified speed ratio. It is sufficient to reject direct pair-field accumulation

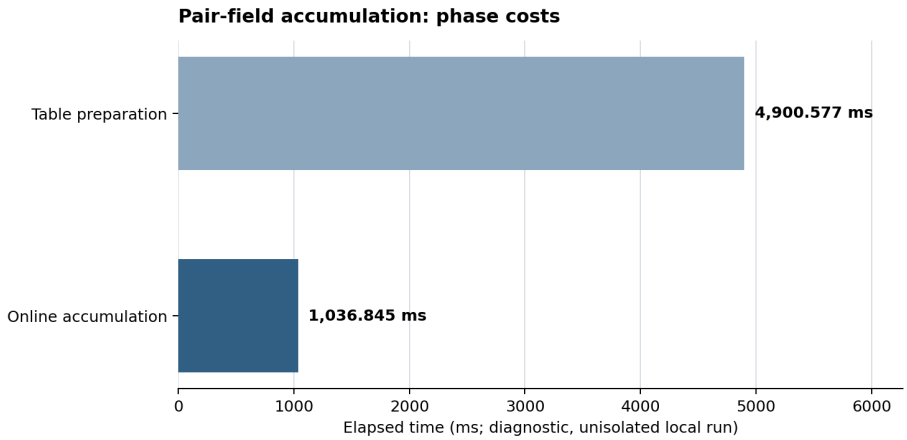


Figure 2: Separate diagnostic phase costs for the audited 12,288-output run

as the next timing candidate, while retaining the measured phase and memory costs for future work. The path remains diagnostic and does not change default scalar multiplication.

Checks and graph status. The receipt auditor was rerun against the committed archive and reproduced its receipt. The pair-field algebraic representation is limited to secp256k1 and the stated subgroup; no new curve, endomorphism, isogeny, candidate route, or measured winner is proposed. The canonical curve/isogeny graph files are not present in this PR's experiment tree, so this report makes no claim that those main-branch graphs were checked. The editable figure source is `pair_accumulation_report_2026-10-19.py`; the generated SVGs and PDF accompany this report.